

PERSPECTIVE

Twenty Years of Evidence-Based Conservation: Progress, Promise, and Future Directions

Tiny forests, huge claims: The evidence gap behind the Miyawaki method for forest restoration

Narkis S. Morales¹  | Ignacio C. Fernández² | Leonardo Durán^{3,4} | Dylan Craven^{5,6}

¹Bioeconomy Science Institute, Lincoln, New Zealand; ²Departamento de Ciencias, Facultad de Artes Liberales, Universidad Adolfo Ibáñez, Santiago, Chile; ³Escuela de Ingeniería Forestal, Facultad de Ciencias, Ingeniería y Tecnología, Universidad Mayor, Santiago, Chile; ⁴Hémera Centro de Observación de la Tierra, Facultad de Ciencias, Universidad Mayor, Santiago, Chile; ⁵GEMA Centre for Genomics, Ecology & Environment, Universidad Mayor, Santiago, Chile and ⁶Data Observatory Foundation, Santiago, Chile

Correspondence

Narkis S. Morales

Email: moralesn@landcareresearch.co.nz

Ignacio C. Fernández

Email: ignacio.fernandez.c@uai.cl

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Abstract

1. Evidence-based methods for restoring ecosystems are essential for successful restoration practices, yet many approaches are adopted without rigorous evaluation. The Miyawaki method, originally developed in the early 1970s, has gained popularity for its purported ability to accelerate biodiversity recovery and ecosystem functioning compared to conventional methods. However, the empirical evidence supporting these claims remains unclear.
2. We conducted a systematic review of scientific literature on the Miyawaki method. Of 51 documents examined, only 41% ($n=21$) provided quantitative assessments, and among these, just 33.3% ($n=7$) included a control and 14.3% ($n=3$) included replication.
3. Reported outcomes offered weak to null evidence for most claimed benefits, including rapid growth, accelerated succession, self-sustainability, cost efficiency, enhanced biodiversity, higher carbon sequestration and increased tree density.
4. *Synthesis and applications.* To scale up restoration effectively, practitioners and policymakers should prioritize methods supported by robust empirical evidence rather than relying on untested claims. Our findings highlight the urgent need for rigorous experimental designs and transparent reporting to guide evidence-based restoration strategies.

KEYWORDS

ecological remediation, ecological restoration, Miyawaki method, urban forests

1 | INTRODUCTION

The growing urgency of the need to restore ecosystems globally (United Nations, 2025) has dramatically expanded the methods available to restoration practitioners and scientists. Among the multitude of methods, the Miyawaki method has emerged as an

increasingly popular strategy for ecosystem restoration (Heuch & Thurman, 2024). Developed in the early 1970s by Dr. Akira Miyawaki for the restoration of tropical forests, the main objective of this method was 'to restore the topsoil on sites and then plant seedlings of the canopy species of the local potential natural forest, based on phytosociological field surveys' (Fujiwara, 1993). In brief,

this method proposed to improve soil quality conditions for the subsequent establishment of shade-intolerant tree species.

The Miyawaki method started to gain popularity following the publication of the book *Restoration of Tropical Ecosystems* (Lieth & Lohmann, 1993), which was based on the conference proceedings of a symposium co-organized by Dr. Miyawaki in 1991. However, this book was later criticized for overemphasizing the role of the Miyawaki method 'as an easy, straightforward, and highly successful method for ecosystem restoration' with no empirical data to support the reported claims (Van der Valk, 1995). Despite an apparent lack of empirical support, the 'Miyawaki method' has grown in popularity globally and is increasingly used for generating small forest patches in urban contexts (Akram et al., 2025).

Here, we first synthesize the published literature about the recommended application of the Miyawaki method for restoring disturbed ecosystems and then carry out a systematic review to (1) identify the main claims about the method's advantages and (2) assess the available empirical data to evaluate these claims.

2 | WHAT ARE THE RECOMMENDED APPLICATIONS OF THE MIYAWAKI METHOD?

Dr. Miyawaki argued that the main advantage of his method is that it does not follow the sequence and timeframes of ecological succession (Walker et al., 2007), implying that it can produce faster results, potentially transforming a bare, disturbed landscape into a quasi-natural forest between 15 and 50 years rather than the 200–300 years that secondary succession may take (Miyawaki, 1993, 1998, 1999). The restoration process described by Dr. Miyawaki closely resembles the remediation or rehabilitation processes described in the current Society for Ecological Restoration restorative continuum (Gann et al., 2019).

Originally, the Miyawaki method was recommended for the restoration of small areas (Miyawaki & Golley, 1993). For such areas, the Miyawaki method recommends planting high densities of tree species that likely occur in the region of the study, that is, from the regional species pool (Jarzyna et al., 2025). Another key element of the Miyawaki method is the use of soil amendments; if topsoil is unavailable, it must be brought to the site or made using a mixture of organic soil and straw. After 3 years, the method recommends suspending active management of restored sites ('no management is the best management'; Miyawaki, 1993). At this point, the expected result is a quasi-natural forest dominated by shade-tolerant tree species after 40–50 years (Bozzano et al., 2014; Miyawaki, 1993, 1999; Miyawaki & Golley, 1993; Van der Valk, 1995). It is important to note that this method does not specify which aspects of a restored site will resemble those of a natural forest, such as species diversity and composition, horizontal and vertical structure or ecosystem functioning (CBD, 2006).

The Miyawaki method was later expanded to include the use of pioneer, intermediate, and late successional tree species to encourage

interspecific competition, which is expected to result in high-density (3–5 individuals per m²), diverse forests (Miyawaki, 1993). High diversity and high-density plantings resemble widely used forest diversity experiments that explore the effects of biodiversity on ecosystem functioning, whose design is intended to accelerate species interactions and reduce space and time constraints typically associated with tree growth trials (Paquette et al., 2018; Tobner et al., 2014). The Miyawaki method also recommends the addition of soils from surrounding forests, as well as organic matter and natural debris to further stimulate tree growth. In terms of management, more recent publications have also incorporated acclimatizing seedlings, using mulch, weeding and irrigation for the first 2 years.

More recently, the Miyawaki method has been adapted for use in urban areas to create small forest patches. This adaptation is usually known as 'tiny forests', 'miniature urban forests', 'micro-forests' or 'Miyawaki urban forests' (Kurian, 2020; Mukherjee et al., 2022). The method used to create these tiny forests closely resembles the original Miyawaki method but differs primarily in the size of the afforested areas, which are typically small, and, in some cases, include the use of non-native species (Kurian, 2020; Martin et al., 2021).

3 | SYSTEMATIC LITERATURE REVIEW

To evaluate the effectiveness of the Miyawaki method, we performed a systematic literature search across four databases: Web of Science (WoS), Scopus, Dimensions, and Google Scholar (Gusenbauer & Haddaway, 2020). We used a search string combining the key term 'Miyawaki' with 'forests', 'methods', 'technique', 'afforestation' and 'restoration', retrieving all types of documents published through 2024 with no language restrictions (Table S1). We found a total of 244 documents, including grey and academic literature (Cadotte et al., 2025). Following the ROSES protocol (Haddaway et al., 2018), we eliminated duplicates, removed conference abstracts because they do not provide full texts, and removed documents if they were about other topics. While we included non-English documents in our search, as excluding non-English languages may bias the results of literature reviews (Nuñez & Tatsuya, 2021), we found 14.4% of documents in languages other than English. We note that using search strings in non-English languages could have increased the number of documents included in our study. After full-text screening, our final database was comprised of a total of 51 full-text documents for empirical analysis (Table S2; Foo et al., 2021). We used these 51 documents to identify the main claims about the Miyawaki method reported in the literature and to assess the level of the empirical evidence supporting them. Only one of these documents (2%) was in a language other than English. Across all documents, we classified the quantitative support for all claims based on the reported results in four categories: strong (>10 documents), partial (<10 documents), weak (<3 documents) and null evidence (zero documents; Table 2).

Among the 51 documents included in our analysis, most were journal articles (69% or $n=35$), followed by conference papers, book chapters, preprints, one thesis and one report. Approximately half

consisted of WoS and Scopus indexed documents (49%), but only 41% ($n=21$ documents) of all documents contained quantitative data for empirical analysis (Figure 1; Table 1). While our systematic review found documents published since 1993, most documents have been published since 2020. A list of data sources used in the study is provided in the Data sources section.

4 | IS THERE EMPIRICAL DATA SUPPORTING THE CLAIMS OF THE MIYAWAKI METHOD?

We found that the two most frequent claims about the Miyawaki method are that it promotes rapid growth (up to ten times faster) and that the forest reaches maturity within a relatively short timespan (20 to 30 years), mentioned in 21 and 20 documents of the 51 documents reviewed, respectively (Table 2). Other recurrent claims include that Miyawaki forests become 'self-sustainable' within 3 years ($n=10$ documents), that the method is cost efficient ($n=9$), enhances biodiversity ($n=8$), increases carbon sequestration ($n=8$) and results in higher tree density ($n=7$) (Table 2).

Despite the frequent claims about the advantages of the Miyawaki method, we found little empirical evidence to support them. Eight documents reported data on forest growth (e.g. height, diameter, canopy cover), partially supporting the claim of rapid growth. However, most of these studies did not account for the potential effects of intensive soil preparation on forest growth. Evidence for carbon sequestration was weak, with only two studies estimating carbon stocks. Although these studies reported higher sequestration during the early stages of restoration, they found no statistically significant differences compared to other methods over longer time periods. Similarly, support for enhanced biodiversity was weak, with only one study directly quantifying it and without quantitative comparisons to alternative approaches. For the remaining claims (i.e. rapid maturity, self-sustainability, cost efficiency and higher density), we found null empirical evidence, as none of the reviewed documents provided data (Table 2).

References included in Table 2: Aarthi et al. 2021 [1], Ahsan 2023 [2], Akbar et al. 2010 [3], Bellarosa & Schirone 1998 [4], Cambria et al. 2024 [5], Chaturvedi et al. 2024 [6], Da & Song 2008 [7], Daou et al. 2024 [8], Ducros 2022 [9], Frattaroli et al. 2017 [10], Fujiwara 2021 [11], Gill 2024 [12], Guo 2018 [13], Hanpattanakit et al. 2022

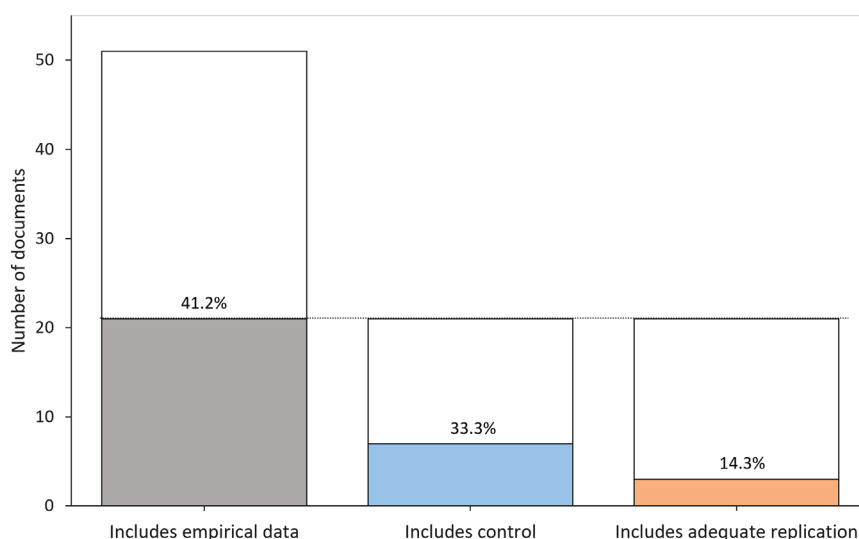


FIGURE 1 Summary of documents included in the systematic review of the Miyawaki method, how many documents included empirical data and, of those, how many included either a control or adequate replication.

TABLE 1 Summary of documents selected in the systematic literature review showing the number of documents by type, indexation and data availability. Documents with data refer to those documents that include quantitative measurements of restoration efforts. Indexation refers to documents in the WoS and Scopus databases.

Document type	Indexed		Not indexed		Total		Total records
	Data	No data	Data	No data	Data	No data	
Journal article	10	8	3	14	13	22	35
Conference paper	2	4	0	1	2	5	7
Book chapter	1	0	2	1	3	1	4
Preprint	0	0	2	1	2	1	3
Thesis	0	0	0	1	0	1	1
Report	0	0	1	0	1	0	1
Total	13	12	8	18	21		51

TABLE 2 Summary of empirical evidence associated with the principal claims of the Miyawaki method. Numbers are associated with documents included in the systematic review (Table S3).

Principal claims in the revised literature	References mentioning claims	Level of evidence	Details	References with empirical evidence
Rapid growth (usually 10 times faster)	1, 2, 8, 9, 10, 12, 15, 21, 26, 27, 28, 29, 30, 37, 39, 41, 43, 44, 47, 48, 50 (n=21)	Partial	Studies report higher growth rates in height, diameter, and canopy for the first several years in Miyawaki forests compared to traditional planting methods. However, most studies do not evaluate the effects of intensive soil management in the Miyawaki method in comparison to other methods.	7, 10, 15, 16, 19, 37, 38, 40 (n=8)
Rapid maturity / accelerated secondary succession (usually in 20-30 years)	1, 2, 7, 8, 10, 12, 15, 16, 17, 18, 26, 27, 28, 29, 39, 40, 42, 43, 47, 48 (n=20)	Null	There are no studies evaluating the maturity of forest established with the Miyawaki method.	- (n=0)
Becomes self-sustainable (usually after 3 years)	1, 2, 6, 12, 15, 17, 24, 26, 30, 39 (n=10)	Null	There are no studies evaluating the capacity of Miyawaki forests to become self-sustainable, nor the time at which this capacity is reached.	- (n=0)
The method is cost efficient	1, 7, 8, 12, 13, 16, 17, 27, 28 (n=9)	Null	There are no studies specifically targeted to evaluate the cost or efficiency of the method, nor a comparison with other alternative methods. However, some documents mention that costs can be higher for the Miyawaki method when compared with other methods.	- (n=0)
Enhanced biodiversity	7, 9, 13, 15, 29, 30, 47, 48 (n=8)	Weak	Studies report biodiversity indicators (e.g. presence of insects and pollinators) in Miyawaki forests. These are not quantitatively compared to those of a traditional forestation method or a natural forest.	38 (n=1)
Enhanced carbon sequestration	5, 9, 14, 17, 26, 27, 30, 41 (n=8)	Weak	Some studies provide data on carbon stocks but not carbon sequestration rates and show no long-term differences between the Miyawaki method and natural or plantation forests.	14, 46 (n=2)
Forest reaches higher density	1, 15, 26, 37, 39, 41, 43 (n=7)	Null	There are no studies evaluating the long-term development and stability of the high initial density promoted with the Miyawaki method.	- (n=0)

[14], Kavana 2023 [15], Kiboi et al. 2015 [16], Kurian 2020 [17], Mandowara 2022 [18], Meguro et al. 2021 [19], Meguro 2023 [20], Mihailova 2022 [21], Miyawaki & Golley 1993 [22], Miyawaki 1993 [23], Parikh & Nazrana 2023 [24], Parikh 2024 [25], Poddar 2021 [26], Qi et al. 2024 [27], Quian et al. 2021 [28], Rajadurai 2024 [29], Ranjan et al. 2015 [30], Ranjan et al. 2016 [31], Ranjan et al. 2017 [32], Resmi et al. 2024a [33], Resmi et al. 2024b [34], Rochard [35], Rots 2019 [36], Roy & Chattejee 2023 [37], Safvan & Swapna 2023 [38], Sandip et al. 2022 [39], Schirone et al. 2021 [40], Sharma et al. 2024 [41], Singh & Saini 2019 [42], Sivalan et al. 2021 [43], Sreelekshmi et al. 2024 [44], Szabo et al. 2021 [45], Szabo et al. 2024 [46], Ullah et al. 2023 [47], Valdes 2023 [48], Wake & Havell 2023 [49], Zhang et al. 2023 [50], Zhao et al. 2024 [51].

5 | DISCUSSION

The Miyawaki method was originally developed for restoring temperate and subtropical forests in Japan and then applied to tropical

forests in Southeast Asia (Miyawaki, 1993). The method has undergone several changes over the years. Its most current definition aims to establish vegetative cover using native species from the area, intensive soil interventions and post-planting management. The Miyawaki method has been widely promoted and applied in different forest ecosystems around the world, largely due to its purported ability to rapidly create diverse, healthy and self-sustainable forests. Although we found frequent claims regarding the supposed advantages of the Miyawaki method in our systematic review, our analysis of the literature suggests that the majority of these claims are not based on empirical evidence.

One possible explanation for the limited empirical support for many claims about the Miyawaki method relates to the types of documents in which these claims are promoted and the date of their publication. In our systematic literature review, we included a range of sources, from peer-reviewed journal articles indexed in major databases (e.g. WoS, Scopus) to non-indexed, non-peer-reviewed documents (e.g. reports, preprints), because including grey literature has been shown to provide important perspectives on conservation and

restoration practices (e.g. Castellanos-Galindo et al., 2011; Cross et al., 2019), particularly if generated by practitioners (Haddaway & Bayliss, 2015). Additionally, the majority of documents published since 2020 may have limited the number of reports containing empirical evidence. While a growing body of literature has shown that language biases frequently affect literature reviews and meta-analyses (e.g. Konno et al., 2020; Nuñez & Tatsuya, 2021), our search found a limited amount of scientific or grey literature published in non-English languages (2%, $n=1$ document). It is possible that using search strings in non-English languages, particularly Japanese, could have increased the number of documents included in our systematic literature review.

We found that only 54% of indexed articles mentioned any of the method's claims (Table 2), compared to 73% of non-indexed documents. Moreover, indexed articles reported on average, only two claims per document, which is 50% fewer than the three claims per document observed in non-indexed sources. This pattern suggests that widely circulated claims about the Miyawaki method may originate primarily from grey literature. Although empirical evidence is scarce, we did find 21 articles that reported quantitative results. However, most of them lack sufficient replication or failed to include direct, quantitative comparisons with a control treatment (Hanpattanakit et al., 2022; Meguro et al., 2021; Szabó et al., 2024). Most of the empirical data included in scientific publications about the Miyawaki method come from restored areas using the method without a control or with limited replication. Of the studies with empirical data, only 14% ($n=7$ documents) of the reviewed publications included a control and 6% ($n=3$) included replication (Figure 1 and Table S2). We believe that the gap in information highlights the need for caution when interpreting these claims and for more rigorous empirical evaluation in the scientific literature.

Gaps become evident when one attempts to find empirical evidence for specific claims. For example, regarding the rapid growth associated with the Miyawaki method (often cited as 10 times faster), the articles we found were based on post-intervention measurements. These growth rates, although higher than those observed in other restoration methods (Frattaroli et al., 2017) are rarely accompanied by direct comparisons with control plots or alternative restoration approaches. This makes it difficult to separate the effects of the site or soil conditions from the influence of the method itself. Nevertheless, high growth rates are not unexpected. It is well established that high tree densities increase competition for light, often resulting in rapid height growth—particularly in diverse tree communities (Safvan & Swapna, 2023). However, the high seedling densities typical of the Miyawaki method can also lead to elevated mortality rates, ranging from 33% to 50% (Da & Song, 2008).

We could not find empirical evidence to support the claims that Miyawaki forests reach maturity within 20–30 years or become self-sustaining after 3 years. The main reason for this is the lack of studies with temporal measurements, recently established experiments or comparisons with natural forests. Without such data, it is difficult to assess whether these outcomes are attributable to the method itself or to other environmental factors.

We did not find evidence that the Miyawaki method is cost efficient. Rather, the available literature suggests that the Miyawaki method is more expensive than other restoration methods (Elliott et al., 2022; Mercurio, 2018; Qi et al., 2024). This is expected as the Miyawaki method is labour intensive, requires a large number of seedlings, soil preparation and amendments, weeding and irrigation. Updating the available data to current dollars, we estimated a cost of \$32,416 USD/ha in India, \$22,542 USD/ha in Sri Lanka, \$14,997 USD/ha in Indonesia, \$12,842 USD/ha in Thailand and between \$287,711 and \$1.3 million USD/ha in the United Kingdom (Dilhan et al., 2009; Elliott et al., 2022; Heuch & Thurman, 2024; Trees outside Woodland, 2024). Moreover, some studies suggest that other restoration methods may be more cost effective than the Miyawaki method. For example, Dilhan et al. (2009) recommend the Popham method, which costs \$210 USD/ha, while Mercurio (2018) proposes modifications to make the Miyawaki method more affordable in Mediterranean ecosystems. Furthermore, Anand and Mulyani (2020) concluded that the Miyawaki method is less appropriate compared to a cheaper alternative involving soil aeration and the creation of depressions to enhance water retention. Despite its high cost, the Miyawaki method is still included among the recommended approaches for large-scale restoration efforts (Heuch & Thurman, 2024; Martin et al., 2021; Qi et al., 2024). Improving the method's cost efficiency requires further research on seedling production, management and monitoring costs, as well as quantitative comparisons with other restoration methods.

Our systematic review clearly shows that most claims regarding the benefits of the Miyawaki method for improving biodiversity or ecosystem processes have not been tested using statistically robust experimental designs, as they do not include controls or sufficient replication (Safvan & Swapna, 2023). Regarding the claims of higher carbon sequestration, the empirical evidence we found was not confirmatory. Since the studies by Szabó et al. (2021) and Hanpattanakit et al. (2022) did not include a control, it is not possible to determine whether the observed results were due to the method itself or influenced by other factors. The high planting density associated with the Miyawaki method is likely to have a positive impact on carbon sequestration, since more trees generally increase carbon storage, particularly when they are the late-successional species typically used with this method. However, as competition increases and mortality rises, similar levels of carbon may eventually be achieved with lower planting densities (e.g. Kremer et al., 2025; Marziliano et al., 2015; Standish et al., 2022).

Previous studies suggest that the high-density planting of the Miyawaki method is inefficient, since simpler techniques, such as pot-hole planting, can achieve greater basal diameters and heights (Hamzah & Majid, 2001). We found that survival percentages of trees planted using the Miyawaki method vary from 62% to 100% (Chaturvedi et al., 2024; Da & Song, 2008; Miyawaki, 1993; Ranjan et al., 2015). However, Hamzah and Majid (2001) used a less intensive site preparation strategy and achieved a survivorship of 91% to 100%, whereas the survivorship of trees planted with the Miyawaki method in the same site was 82%. Experimental forest ecosystems

often employ high tree densities to accelerate ecological processes such as spatial complementarity for light (e.g. Williams et al., 2017). However, the primary objectives of these studies are ecological—typically testing the direction and magnitude of diversity effects on ecosystem functioning (Paquette et al., 2018)—rather than applied, as they usually rely on standardized tree densities and exclude alternative restoration approaches. Importantly, the relative influence of ecological processes can shift with environmental context; for instance, facilitation may have stronger effects on community assembly and ecosystem functioning than competition under conditions of greater environmental stress (Maestre et al., 2021). This suggests that the design and success of restoration projects could be improved by more explicitly integrating ecological theory (Lindenmayer, 2020). Accordingly, the effectiveness of the Miyawaki method—or other restoration strategies—may depend, at least in part, on which ecological processes dominate in each ecosystem.

We commend the public sector's generally positive attitude towards financing and implementing efforts to address key environmental issues. However, we argue that efforts to restore ecosystems—especially when taxpayer money is used—should take into account empirical evidence to ensure the efficient use of resources (e.g. Kimball et al., 2015). The success (or failure) of restoration efforts also transcends economic costs, as the social dimensions of restoration efforts are also crucial (e.g. Metcalf et al., 2015). If the public finds out that restoration methods such as the Miyawaki method are not supported by empirical data, they may be less willing to participate in or financially support future initiatives. It is therefore crucial to thoroughly—and transparently—evaluate the socio-ecological effectiveness and cost efficiency of multiple restoration methods before their broader implementation.

Our systematic review reveals that there is still a lack of empirically driven assessments of the ecological effects and potential socio-economic benefits of the Miyawaki method (Schirone et al., 2011). We urge restoration scientists and practitioners to exercise greater caution when implementing methods, whatever they may be, that make claims about their effectiveness without empirical evidence.

AUTHOR CONTRIBUTIONS

Narkis S. Morales conceived and presented the original idea. Narkis S. Morales, Ignacio C. Fernández, Leonardo Durán and Dylan Craven contributed to the study conception and design. Ignacio C. Fernández did the literature search. Narkis S. Morales wrote the first draft. All authors commented on the manuscript draft and contributed to writing the final manuscript.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

A list of data sources used in the study is provided in the Data sources section.

ORCID

Narkis S. Morales  <https://orcid.org/0000-0002-3942-1916>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. Search string and custom options used in each database for the systematic review. All searches and records collection were done on 15 July 2025.

Table S2. Studies that include a comparison between the 'Miyawaki method' at different sites or with other restoration methods. References details are provided in Table S3.

Table S3. Studies included in the systematic review. SC=Scopus, WOS=Web of Science, DM = Dimensions, GSC=Google Scholar, Index = SC or WOS indexed, QD=reference included quantitative data, ID=reference identifier used in Table 2.

Figure S1. ROSES diagram showing the steps and screening criteria to produce the literature database used for the analysis.

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